

the Systems tab of the Create panel. Bones are drawn by left-clicking to set the anchor point, moving the mouse to set the length, and then left-clicking again to set the end point of the bone and the start of the next bone. Right-clicking ends the operation. When multiple bones are drawn, each new bone is created as a child of the preceding bone, forming a hierarchy. Bone chains can be created with or without inverse kinematics (IK) enabled. When IK is enabled, the IK chain is drawn from the first to the last bone in the chain. The Create bones panel Bones can also have fins. These are protrusions on the sides of the bone that can be used to more precisely fit the bone to the mesh.

6.1.2 Editing Bones

Bones have a Modify panel where the size of the bone as well as the size and number of fins can be changed. In addition, 3ds Max provides a Bone Tools window (Animation > Bone Tools) to further refine the skeleton. Removing Bones leaves the hierarchy intact, but deleting them breaks the hierarchy. The refine option on the other hand lets you split the bone in two.

6.2 Inverse Kinematics

By default, bones in 3ds Max move strictly by rotation. This is called forward kinematics, because a series of bones is manipulated from the root bone forward. If a character picks up a cup of coffee, the bones are posed starting at the shoulder and moving forward to the fingertips. Forward kinematics is great for most motions, but it can pose a serious problem whenever a character needs to keep one part of the body stable while the other moves, such as keeping a foot on the ground during a walk or a run.

To overcome this limitation, inverse kinematics (IK) can be used. Inverse kinematics automatically rotates a chain of bones so that the end points can be positioned by using translation instead of rotation. This is perfect for a character's legs, but also can be used in other areas, such as arms. If a character climbs a ladder, for example, the hands have to remain stable on the rungs while the shoulders move. Inverse kinematics allows you to position the bones simply by translating the IK handle to the desired location, using one step. The



Edit bones to adjust the posture of your character